# New Calculations of Numerical Invariants Related to the Big Finitistic Dimension

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# Table of Contents

1 Introduction

2 Some New Calculations

3 Some Open Questions

# Introduction

Throughout, let $\mathbf{k}$ be an algebraically complete field and $A = \mathbf{k}Q/I$ be a finite-dimensional quiver $\mathbf{k}$-algebra with relations.

## Conjecture (Bass, 1960s)

The **big finitistic dimension conjecture (big FDC)** claims that the **big finitistic dimension** of $A$, denoted by $\mathrm{FinDim}(A)$:

$$\mathrm{FinDim}(A) := \sup\{\mathrm{pd}(M) : M \in \mathrm{Mod}\, A \text{ such that } \mathrm{pd}(M) < \infty\}$$

is finite for all $A$.

I will happily copy the "Motivation" from Odysseas Giatagantzidis, thanks :) .

# Remarks on Syzygies

Given a finitely presented (right) $A$-module $M \in \operatorname{mod} A$,

**Definition**

Denote $\Omega^n(M)$ as the $n^{th}$ syzygy of $M$. We will see $\Omega$ as a functor from the projective stable category $\underline{\operatorname{mod} A}$ to itself. We also denote the left adjoint to $\Omega$ by $\Sigma$.

$\Sigma$ can be calculated by $\operatorname{Tr}\Omega\operatorname{Tr}$, where $\operatorname{Tr}$ is the transpose of the module $M$.

# Delooping Level

Given a finitely presented $M \in \operatorname{mod} A$:

## Definition (Gélinas, 2022)

The **delooping level** of $M$ is the least integer $n$, such that there exists $N \in \operatorname{mod} A$ with $\Omega^n(M)$ as a summand of $\Omega^{n+1}(N)$. We write $\operatorname{dell}(M) = n$ when such $n$ exists.

## Definition (Gélinas, 2022)

The delooping level of an *algebra* is the supremum of delooping level over all simple modules of $A$.

## Theorem (Gélinas, 2022)

*For any algebra* $A$, $\operatorname{FinDim}(A^{op}) \leq \operatorname{dell}(A)$.

# New Invariants

## Definition (Guo and Igusa, 2025)

The $k$**-delooping level** of $M$ is the least integer $n$ such that there exists module $N$ with $\Omega^n(M)$ as a summand of $\Omega^{n+k}(N)$. We write $k\text{-dell}(M) = n$ when such $n$ exists.

## Definition (Guo and Igusa, 2025)

The **derived delooping level** of an $A$-module $M$ is:

$$\operatorname{ddell} M = \inf\{m \in \mathbb{N} \mid \exists n \leq m \text{ and an exact sequence in } \operatorname{mod} A$$
$$0 \to C_n \to C_{n-1} \to \cdots \to C_1 \to C_0 \to M \to 0$$
$$\text{where } (i+1)\text{-dell}\, C_i \leq m - i, i = 0, 1, \ldots, n\}$$

# What do we know about these invariants?

**Theorem (Kershaw and Rickard, 2024)**

*There is an Algebra $A$ where* $\operatorname{dell}(A) = \infty$.

**Theorem (Guo and Igusa, 2025)**

$\operatorname{FinDim}(A^{op}) \leq \operatorname{ddell}(A) \leq k\text{-}\operatorname{dell}(A)$ *for all* $k$.

**Theorem (Chen, 2026)**

$\operatorname{dell}(A) < \infty$ *is* ***not*** *preserved under derived equivalence.*

# Table of Contents

1 Introduction

2 Some New Calculations

3 Some Open Questions

# New Calculations

Barrios, Lanzilotta, Mata considered a collection of quiver algebras depending on two positive integer $k < n$.

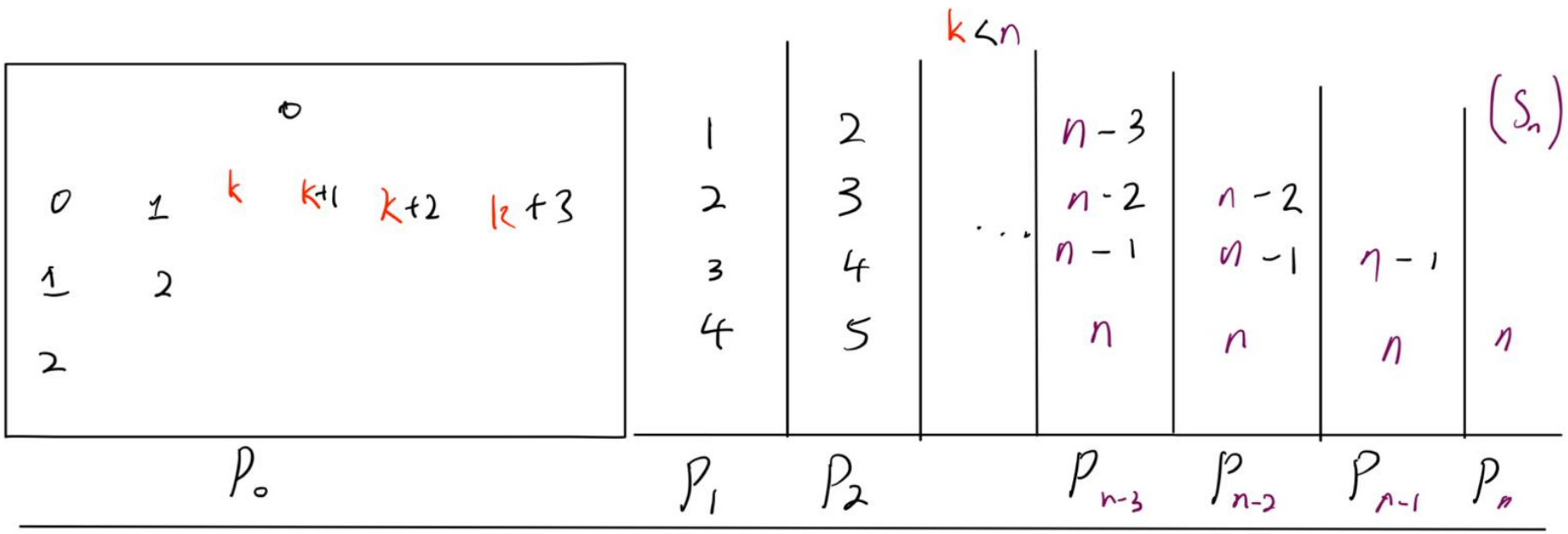
**Theorem (L.)**

*For this collection of algebras, we have*
$\mathrm{ddell}(A) = \mathrm{FinDim}(A^{op}) = O(k)$ *and* $1\text{-}\mathrm{dell}(A) = O(n)$.

$O(n)$ just means $\left\lfloor \frac{n+1}{2} \right\rfloor$ or something like that, I am lazy.

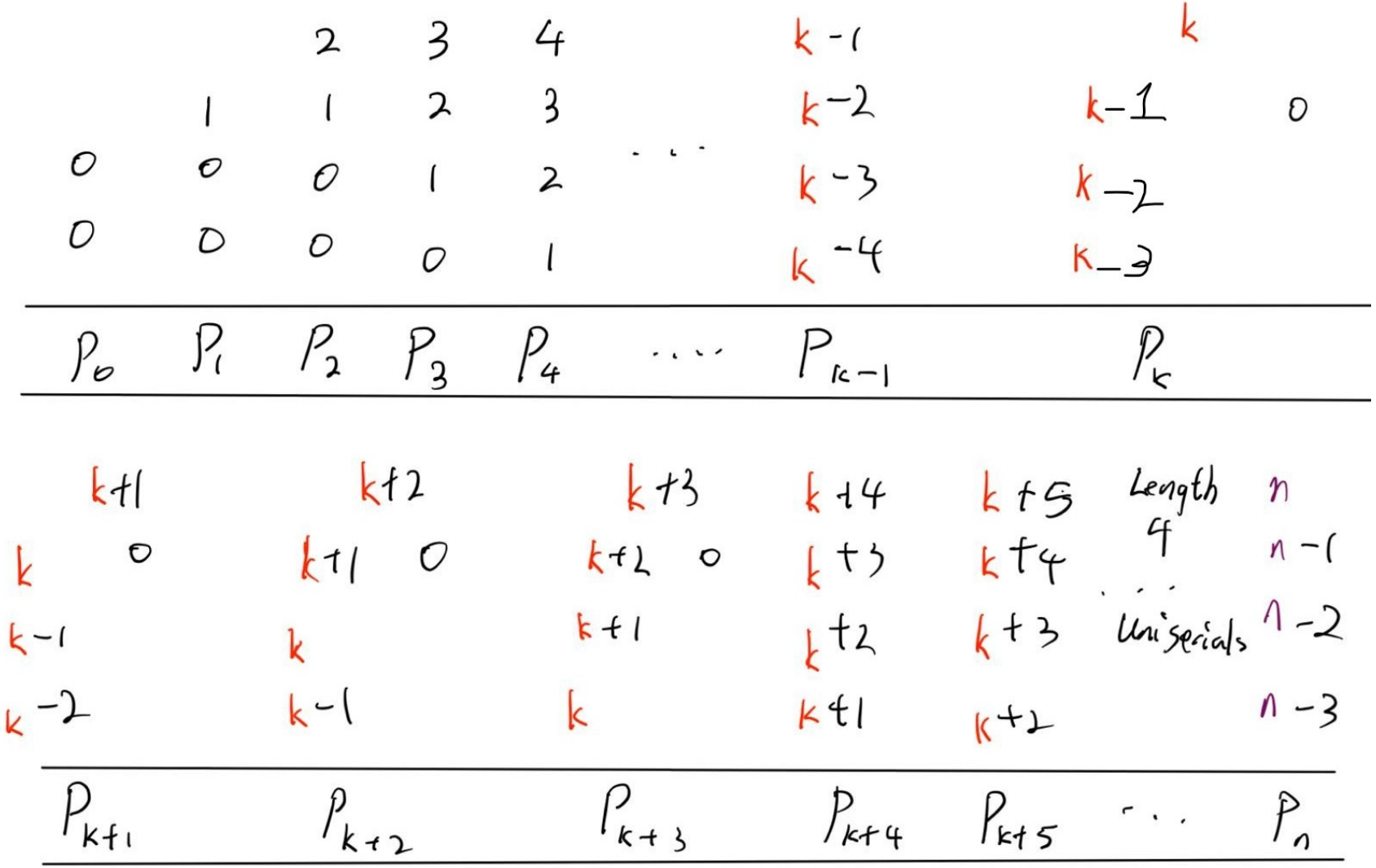
# The Barrios-Lanzilotta-Mata Example

The example is a quiver algebra with monomial relations.
The Projective Indecomposables of $A$:



# The Barrios-Lanzilotta-Mata Example (2)

The Projective Indecomposables of $A^{op}$:



# Computing dell and ddell (1)

Informally, justifying $\mathrm{FinDim}(A^{op}) = \mathrm{ddell}(A)$ breaks down to a few steps:

- Calculating $\mathrm{FinDim}(A^{op}) =: f$ using the result by Huisgen-Zimmermann.
- Arguing $\mathrm{ddell}(S_0)$ is the derived delooping level of the *algebra.*
- Write down short exact sequences of length $f$ with $S_0$ as the last term and check every term has the required $i$-delooping level.

# Computing FinDim

The algebra is monomial. So $\mathrm{FinDim}$ can be calculated with high accuracy due to a result of Huisgen-Zimmermann.

**Theorem (Huisgen-Zimmermann, 1991)**

*Let $A$ be a monomial algebra, then consider the finite collection*

$$\mathcal{T} := \{qA : q \text{ is a non-zero path of length} \geq 1 \text{ and } \mathrm{pd}(qA) < \infty\}$$

*Then let*

$$s = \begin{cases} -1 & \text{if } \mathcal{T} = \emptyset \\ \max\{\mathrm{pd}(t) : t \in \mathcal{T}\} & \text{otherwise} \end{cases}$$

*Then* $\mathrm{FinDim}\, A = s + 1$ *or* $s + 2$.

# Computing dell and ddell (2)

**Lemma**

*If a module $M$ is a submodule of a projective module $P$, then* $\mathrm{dell}(M) = 0$.

**Proof.**

$M = \Omega(P/M)$. □

**Theorem (Gélinas, 2022)**

$\mathrm{dell}(M) \leq n \Leftrightarrow \Omega^n(M)$ *is a summand of* $\Omega^{n+1}\Sigma^{n+1}\Omega^n(M)$

# Computing dell and ddell (3)

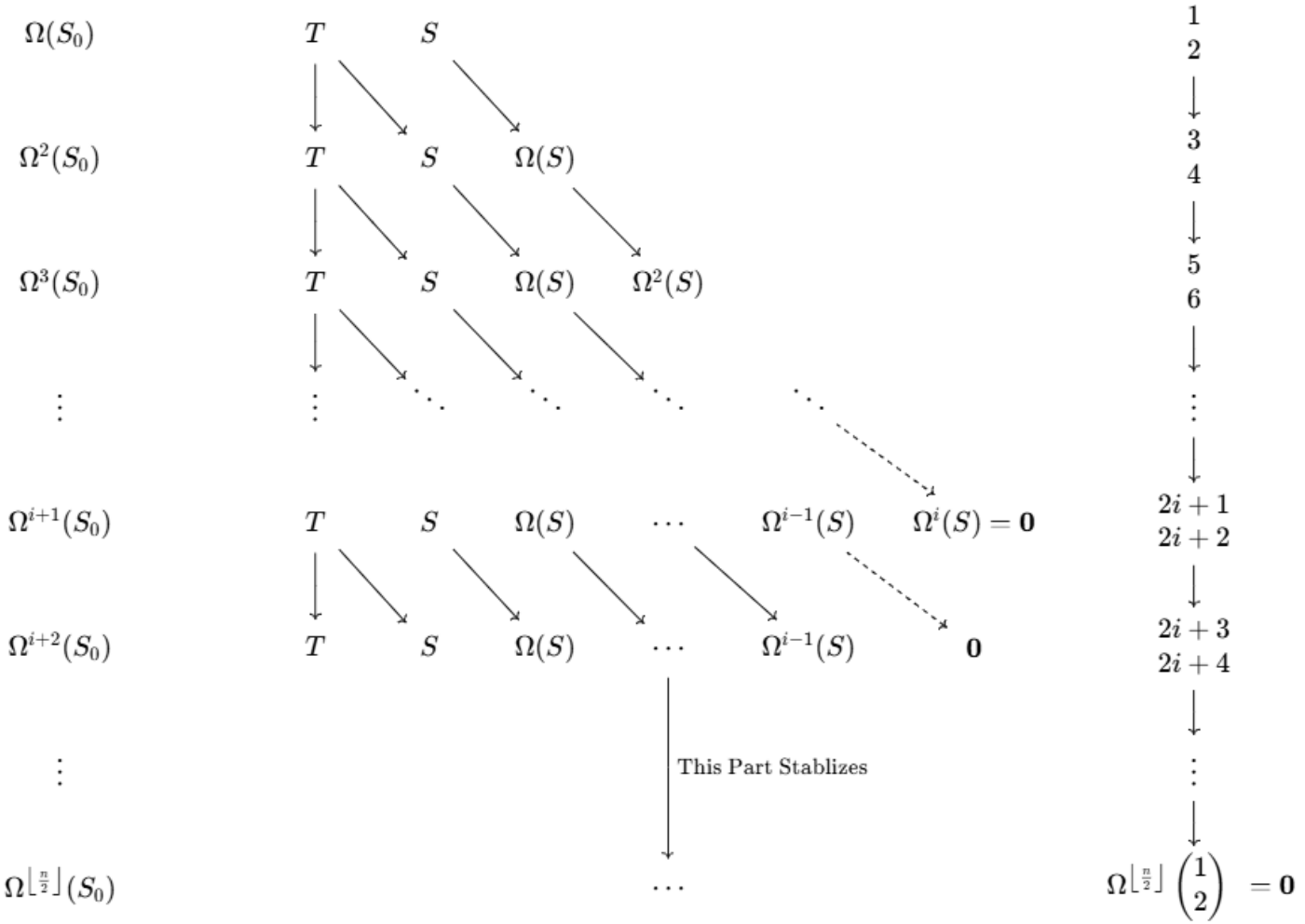



Figure: A picture calculating the syzygies of $S_0$.

# Table of Contents

1 Introduction

2 Some New Calculations

3 Some Open Questions

# Open Questions

- Does there exist $A$ with $\mathrm{ddell}(A) \neq \mathrm{FinDim}(A^{op})$?
- Is $\mathrm{ddell} < \infty$ a derived invariant property?
- I personally know no efficient algorithms for calculating $\mathrm{ddell}(A)$ other than lower bounding it by $\mathrm{FinDim}(A^{op})$ and "guessing" short exact sequences, even for monomial algebras.
- Can algorithms of finding $\mathrm{dell}$ or $\mathrm{ddell}$ (if exists) be generalized to different constructions of algebras? (e.g Special Biserial Algebras? Matrix Algebras?)

Thank you!